Trust Index Notes

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Abstract

Working notes on the definition of AI-judge trustworthiness.

1 Notation

Table 1: Notation.

Symbol	Meaning
Items and agents	
x	candidate artifact
y^*	gold / reference artifact
$\hat{y},\ \hat{y}(c)$	agent output on input c
A	set of agents
$a \in A$	an agent/solver
L_A	agent/solver list (the quantified population in TW_m)
B	set of benchmarks
$b \in B$	a benchmark
Judges	
AJ	set of AI-judge configurations
$aj \in AJ$	judge configuration (model, prompt, K_{aj}, aggregation)
$\theta_j,\ \theta_{aj}$	parameters of aj (prompt, weights, feedback, …)
TJ	set of target judges
$tj \in TJ$	a target judge
$t_j,\ H,\ h_j$	target / human judge (e.g. if $t_j = h_j$)
$\pi,\ \Pi$	rater persona; set of personas
t_j	target persona (alias of the persona on h_j)
a_j	persona realized by configuration aj
K_{aj}	number of judge repeats
K_a	number of agent generations
K_h	number of human passes
Scores and distributions	
$s_{aj}(x, y^*, \hat{y})$	score of aj on a triple; range $[0, 1]$
$\hat{s}_{aj}$	finite-sample estimate of s_{aj}
$s^*_H,\ \hat{S}^*_H$	target/human score and its finite-sample version
$p_{\mathrm{AJ}}(s \mid \cdot)$	pdf/pmf of the AI-judge score
$p_T(s \mid \cdot)$	pdf/pmf of the target-judge score
$\hat{p}_{\mathrm{AJ}}$	finite-sample estimate of p_{AJ}
$\hat{p}_T$	finite-sample estimate of p_T
Trust objects	
$\mathrm{TW}_m(aj)$	trustworthiness of aj at matching level m (construct; a predicate)
$\mathrm{TI}_m(aj)$	Trust Index of aj at matching level m (criterion; needs labels)
$\mathrm{PTI}_p(aj)$	Proxy Trust Index of aj on property set p (instrument; label-free)
$m,\ m_0, m_1, m_2$	matching-criterion index and its ladder
$p \subseteq \{1, \ldots, K\}$	property subset used in PTI_p
P_k	score of behavioral property k
GM	geometric mean
$\approx_\Pi$	persona precondition (aj conditioned toward h_j; equivalently $t_j = a_j$)
$\approx_m$	match under criterion m

Notation note. $\hat{p}_T$ denotes the target-judge pdf/pmf; the hat indicates that the estimate is finite-sample. We write $\hat{p}_T$ (or $\hat{p}_H$ when the target is human) and $\hat{p}_{\mathrm{AJ}}$ for the AI-judge side. In short: we are comparing a finite-sample AI judge against a finite-sample human target.

2 Behaviors and agents

Definition 1 (Universally trustworthy AI judge). Let B be a set of benchmarks and A a set of agents. An AI judge $aj \in AJ$ is universally trustworthy if, for every benchmark $b \in B$ and every agent/solver $a \in A$, aj distributionally matches the corresponding target judge ($tj \in TJ$) (e.g., a human, a persona, a strict rubric, a lenient rubric, etc.):

$$\forall b \in B,\ \forall a \in A, \quad \text{if } t_j = a_j \quad \Rightarrow \quad \hat{p}_{\mathrm{AJ}}\big(s \mid x,\, \hat{y}(c),\, y^*,\, \theta_j\big) \;=\; \hat{p}_{TJ}\big(s \mid x,\, \hat{y}(c),\, y^*,\, \theta_* + t_j\big),$$

where equality is e.g. $\mathrm{KL} = 0$.

Remark. The precondition $t_j = a_j$ (persona / target match) is required. If, for example, we set $t_j =$ strict but the judge a_j has parameters/configuration $e.g., \theta_{a_j}$ (prompt, weights, etc.) that to a lenient judge, then the definition is unsatisfiable.

Remark. The literature often uses preference as a synonym for human judgment. The same word is also used for the objective, so preference is the more general term: it includes trustworthiness. Once a frame of reference is chosen (an objective or not; e.g. a persona), there is no leftover contrast of the form "this is trustworthy but I do not prefer it," or the reverse.